

TEJASWI SREERANGAM

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EDUCATION

University of Illinois Chicago

Master of Science, Electrical and Computer Engineering

Aug 2024 - May 2026

Savitribai Phule Pune University

Bachelor of Engineering, Electronics and Telecommunications

Jul 2019 - May 2023

PROFESSIONAL EXPERIENCE

InnoIndustry

Volunteer Verification Engineer

Aug 2026 - Present

United States

- Contribute to SystemVerilog verification across 9 RTL blocks for 2 FPGA/RTL projects, Hybrid Beamforming and LDPC, covering datapath, memory, FSM/control, reset, and interfaces.
- Verified a 64x24-bit complex 1W2R memory block, validating 1 synchronous write and 2 combinational read paths through simulation and waveform analysis.
- Develop self-checking tests and MATLAB golden-model validation workflows, collaborating with RTL and algorithm engineers on debugging and module integration.

Johnson Controls

Graduate Design Engineer

Oct 2023 - Jun 2024

India

- Performed hardware/software integration validation for embedded sensor and controller modules across 20 multi-story floor plans, executing interface and functional checks against system requirements.
- Analyzed 200+ engineering records to trace failure conditions and corrective actions with design engineers, reducing rework by 15%.
- Developed Python automation for an 8-field engineering traceability workflow, reducing technical-record lookup time by 25%.

PROJECTS

Memory-Mapped DMA Controller - RTL and UVM Verification

Sep 2025 - Jan 2026

- Designed a 32-bit DMA controller with 8 MMIO registers, programmable address generation, transfer FSM, and interrupt logic supporting 1-1,024-word transfers.
- Built a UVM environment to verify 8 registers and 3 functional areas, configuration, data transfer, and interrupts, using sequences, driver, monitor, scoreboard, SVA, and coverage.
- Executed 1,000 transactions, achieving 95% functional coverage and resolving 6 protocol/control defects through regression and waveform debugging.

FPGA-Validated 5G Receiver RTL Blocks

Aug 2025 - Oct 2025

- Designed Verilog RTL for 3 fixed-point 5G processing blocks: LDPC encoder/decoder, MMSE equalization, and CORDIC-based QR decomposition.
- Verified 3 behavior classes, datapath, control, and reset through simulation and waveform analysis, reaching 95% functional coverage.
- Performed Vivado synthesis and timing analysis to achieve 250 MHz, reducing implementation area by 20% and latency by 30%.

Architectural Simulation & CPU Performance Evaluation

Oct 2024 - Dec 2024

- Built a SimpleScalar/PISA environment and evaluated SPEC CPU2006 workloads across 500M instructions, analyzing IPC/CPI and out-of-order execution behavior.
- Explored 5 pipeline widths (1/2/4/8/16) by varying 4 processor resources, fetch, decode, issue, and commit using a cost model.
- Compared IPC/CPI and resource-cost trade-offs across 5 configurations to identify pipeline-scaling trends and performance bottlenecks.

SKILLS

- **ASIC/SoC Verification:** SystemVerilog, UVM, SVA, Constrained-random and directed testing, Functional coverage, Coverage closure, Scoreboards, Monitors, Sequences, Checkers, Regression testing, Waveform debugging, Golden-model verification, RTL testbench development
- **RTL & Digital Design:** Verilog, RTL design, FSMs, Datapaths, MMIO, Clock/Reset logic, Fixed-Point arithmetic, FPGA Prototyping, Registers/FIFO design
- **Programming & Automation:** Python, Bash, C, C++, MATLAB, Linux, Git
- **EDA & Simulation:** Vivado/XSim, Verilator, GTKWave, FPGA Synthesis, TimingAnalysis
- **Computer Architecture & Interfaces:** Pipelining, Out-of-Order execution, IPC/CPI, Memory-Mapped interfaces, CAN, UART, SPI, I2C